

Logistic regression assumptions and diagnostics

Motivating example: Dengue data

Data: Data on 5720 Vietnamese children, admitted to the hospital with possible dengue fever. Variables include:

- + *Sex*: patient's sex (female or male)
- + *Age*: patient's age (in years)
- + *WBC*: white blood cell count
- + *PLT*: platelet count
- + other diagnostic variables...
- + *Dengue*: whether the patient has dengue (0 = no, 1 = yes)

Previously: Logistic regression model

Y_i = dengue status (0 = negative, 1 = positive)

$$Y_i \sim \text{Bernoulli}(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 WBC_i$$

What assumptions does this logistic regression model make?
How should we assess these assumptions? Discuss with your neighbor for 2--3 minutes, then we will discuss as a group.

Logistic regression assumptions

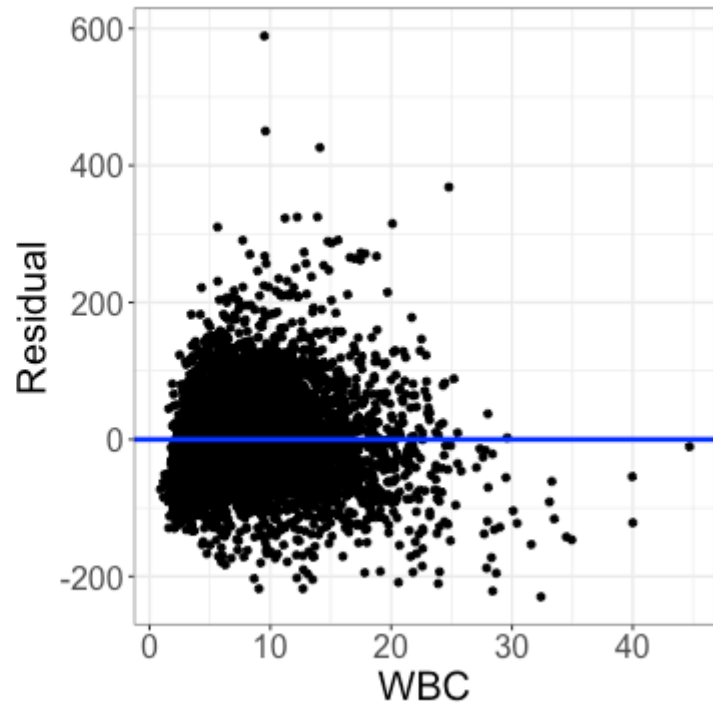
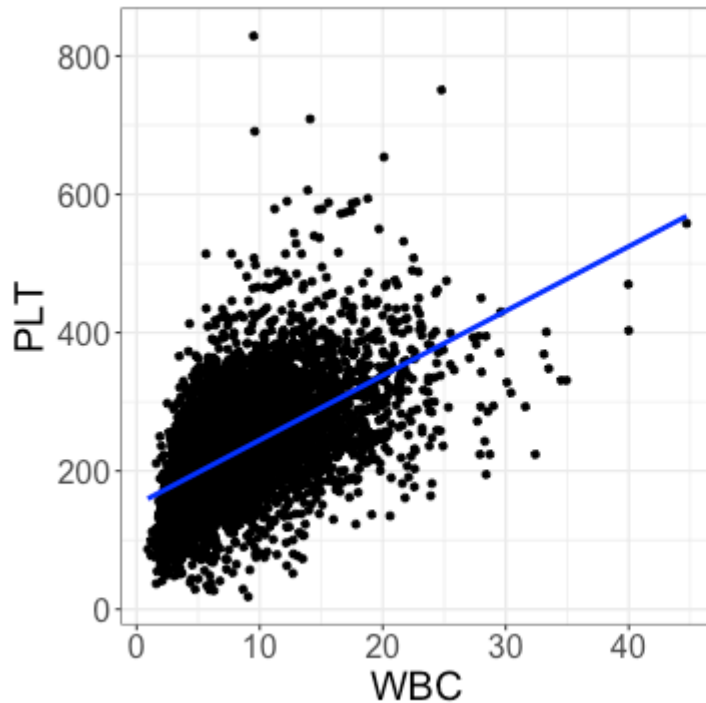
- independence (observations are independent) (needed for MLE)
- binary response variable
- no errors in the data
- randomness (random sample, or data come from a random process)
- Don't need mean = 0 or constant variance assumptions
(b/c don't have $\epsilon \sim N(0, \sigma^2)$ in logistic model)
- Shape: log-odds are a linear function of explanatory variables

Assess

- randomness & independence: think about data generating process
- response: check the data!
- shape: try to plot log odds, some kind of residual plot?
- multicollinearity (later this week)

Residual plots for linear regression

$$PLT_i = \beta_0 + \beta_1 WBC_i + \varepsilon_i$$



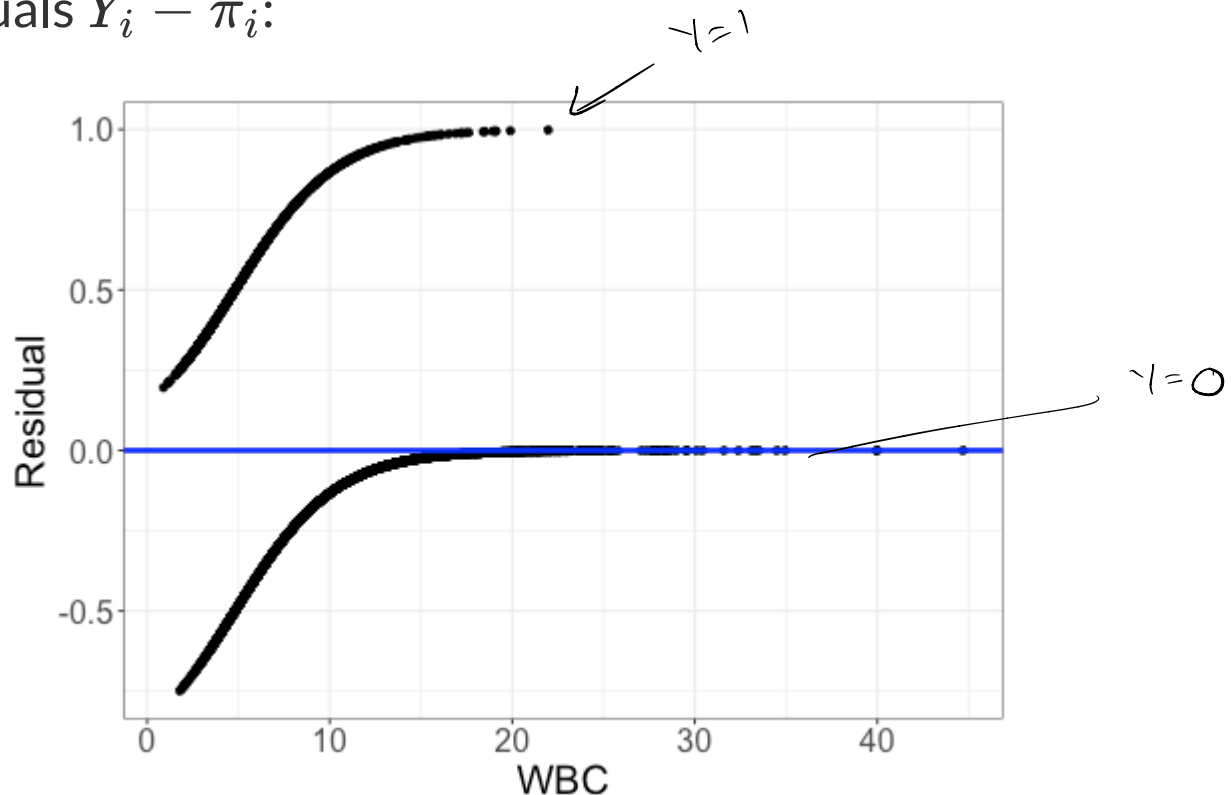
residual: $PLT - \hat{PLT}$

Don't use raw residuals for logistic regression

Fitted model: $\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = 1.737 - 0.361 WBC_i$

Raw residuals $Y_i - \hat{\pi}_i$:

$Y_i \in \{0, 1\}$



Assessing shape with empirical logit plots

Example: Putting data. Interested in the relationship between the length of a putt, and whether it was made:

$$Y_i \sim \text{Bernoulli}(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{Length}_i$$

← want to assess shape assumption

Length	3	4	5	6	7
Number of successes	84	88	61	61	44
Number of failures	17	31	47	64	90
Total	101	119	108	125	134

idea: estimate $\log\left(\frac{\hat{\pi}}{1-\hat{\pi}}\right)$ for each length, and plot against length, check if linear

Empirical logits

Step 1: estimate the probability of success for each length of putt

Length	3	4	5	6	7
Number of successes	84	88	61	61	44
Number of failures	17	31	47	64	90
Total	101	119	108	125	134
Probability of success $\hat{\pi}$	0.832	0.739	0.565	0.488	0.328

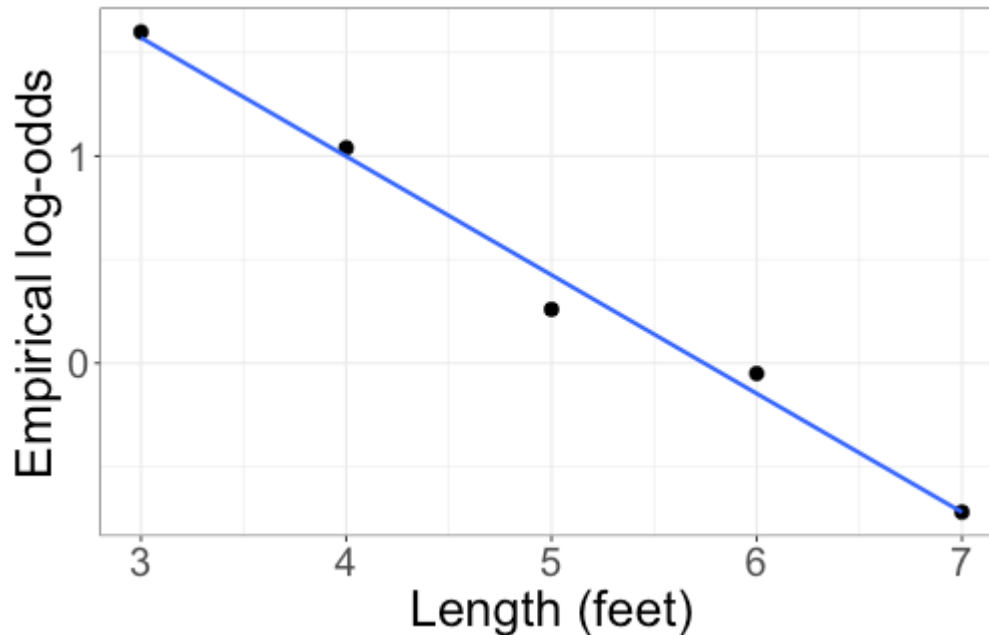
Empirical logits

Step 2: convert empirical probabilities to empirical log odds

Length	3	4	5	6	7
Number of successes	84	88	61	61	44
Number of failures	17	31	47	64	90
Total	101	119	108	125	134
Probability of success $\hat{\pi}$	0.832	0.739	0.565	0.488	0.328
Odds $\frac{\hat{\pi}}{1 - \hat{\pi}}$	4.941	2.839	1.298	0.953	0.489
Log-odds $\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right)$	1.60	1.04	0.26	-0.05	-0.72

Empirical logits

Step 3: plot empirical log-odds against predictor, and add a least-squares line



*Shape looks
pretty
good!*

Does it seem reasonable that the log-odds are a linear function of length?

Back to the dengue data...

WBC	0.90	1.15	1.23	1.25	1.54	1.58	...
Dengue = 0	0	0	0	0	0	0	...
Dengue = 1	1	2	1	1	3	1	...

What problem do I run into?

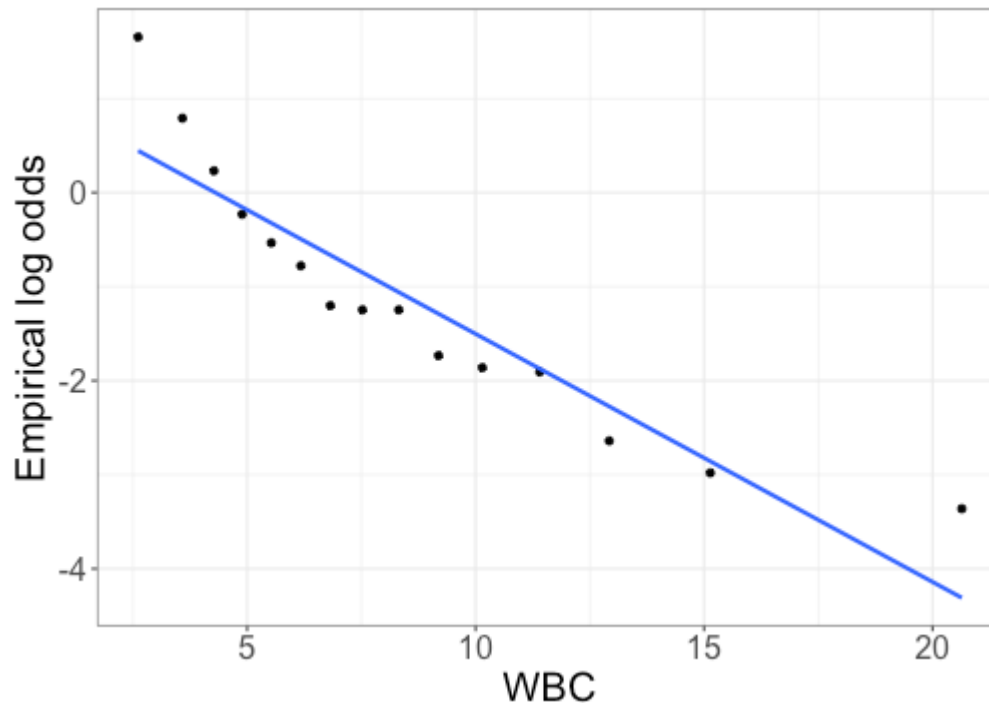
• Many different values of WBC, few observations at each value

$\log(1)$, $\log(1)$, ...

Possible values for log-odds:

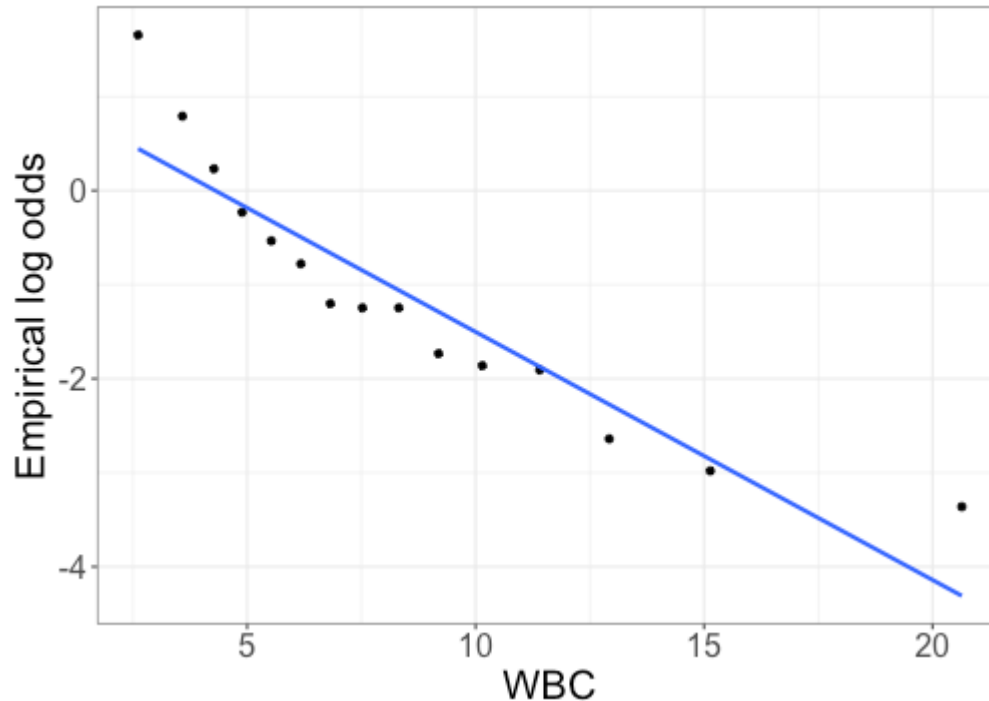
$\log(1)$, $\log(0)$, $\log(\frac{1}{2})$, ...

Binned empirical logit plots



- 1) specify # of bins (hopefully at least 8-10, but depends on data size)
- 2) Divide data into bins based on WBC
(e.g. $WBC \in [0, 1]$, $WBC \in (1, 2]$, ... $[4, 5]$, ...)
- 3) Calculate empirical log odds in each bin

Binned empirical logit plots



· pretty close
to linear

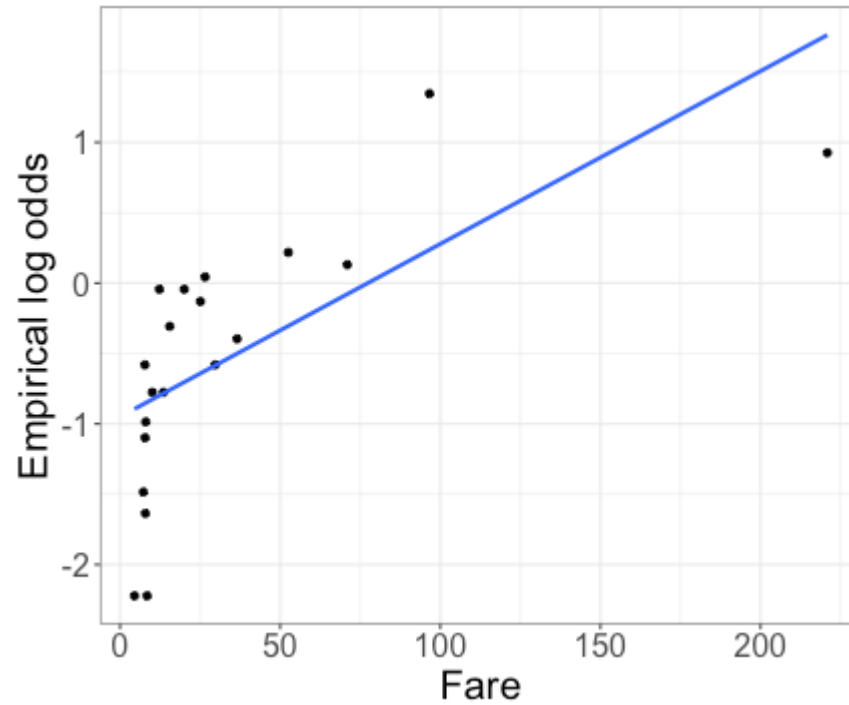
Does it seem reasonable that the log-odds are a linear function of WBC?

Class activity, Part I

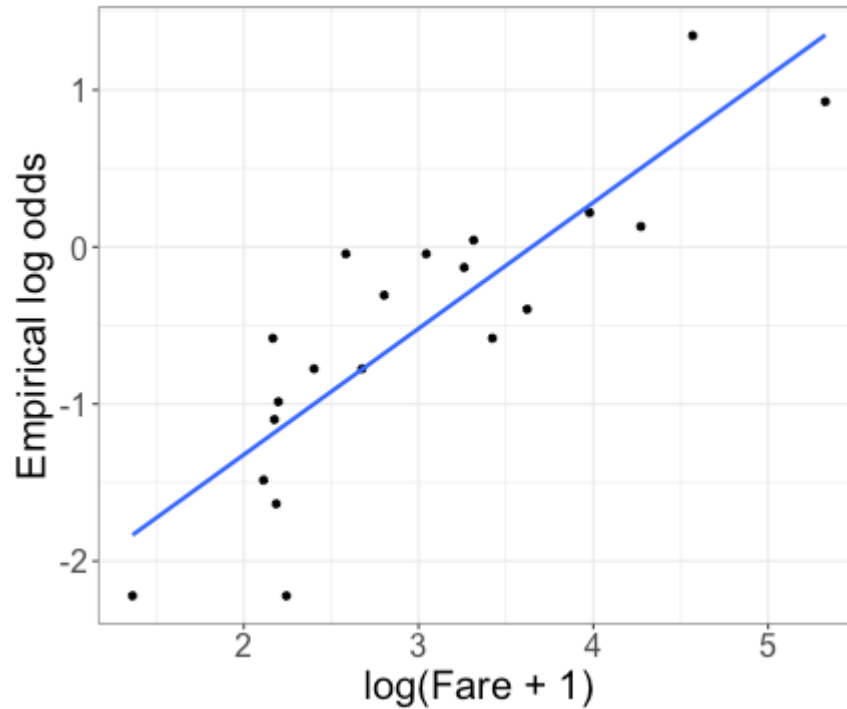
https://sta214-f25.github.io/class_activities/ca_07.html

(Don't need to submit this one)

Class activity



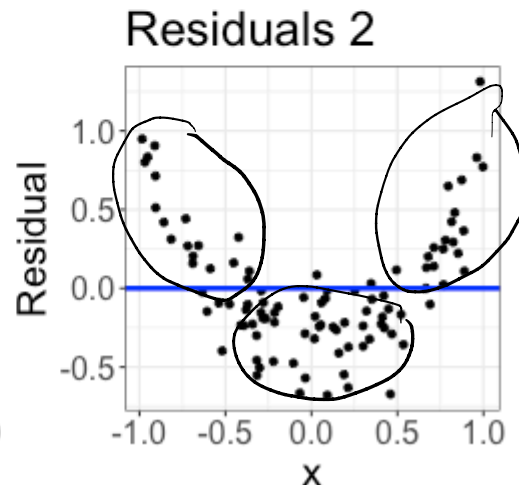
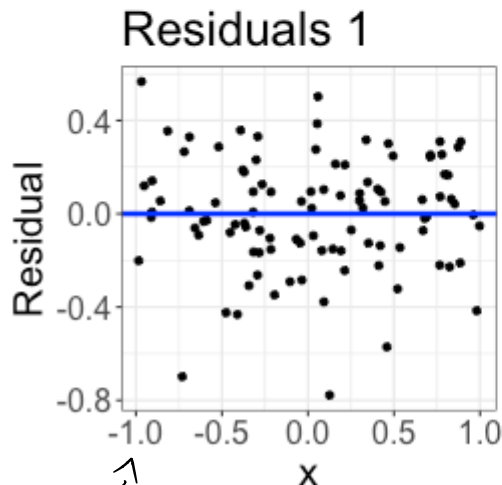
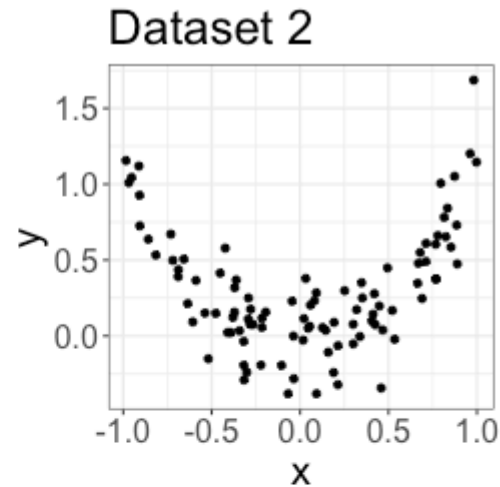
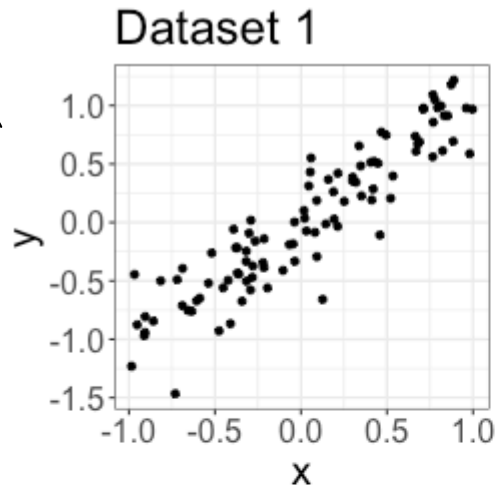
Class activity



$\log(\text{Fare})$ doesn't work $\rightarrow \log(0)$ is undefined / infinite

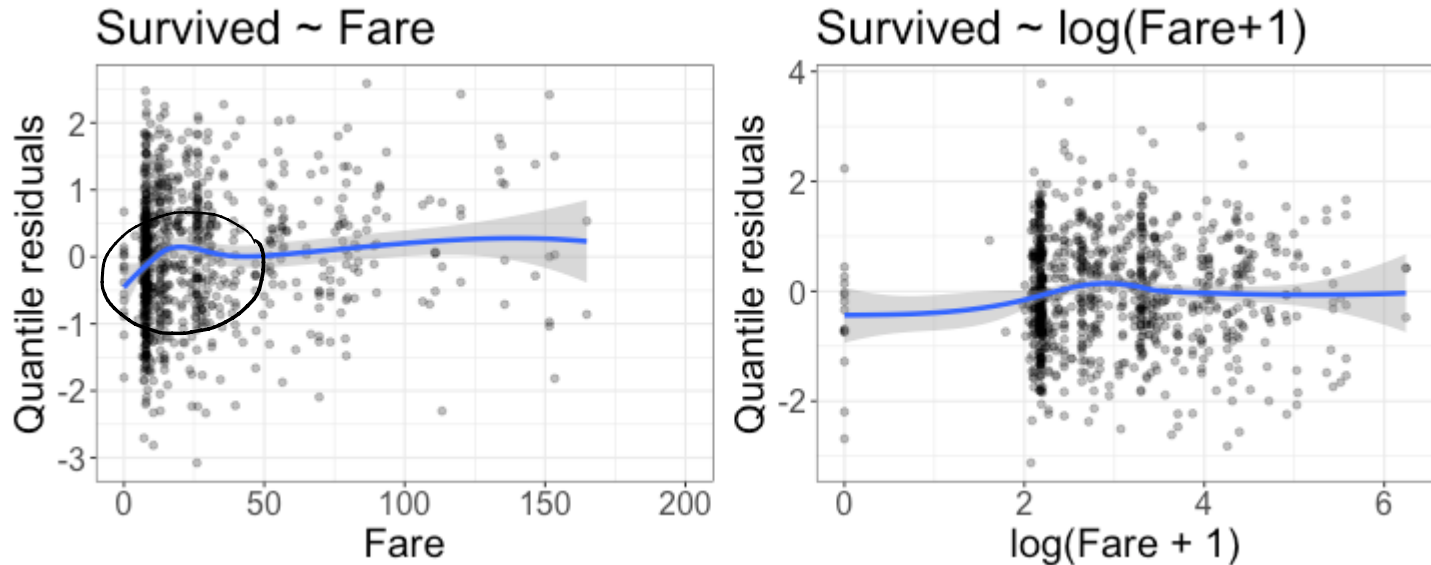
Why residuals in linear regression are nice

- residual for every observation
- use residual plot to identify non-linearity



random scatter around horizontal line

Quantile residuals for logistic regression



- quantile residual: a different residual (not $\hat{y} - y$) that behaves like residuals in a linear model
- Gives us a way to make residual plots for logistic regression models

Class activity, Part II

https://sta214-f25.github.io/class_activities/ca_07.html

(Don't need to submit this one)